

RESEARCH ARTICLE

Improving ETF Prediction Through Sentiment Analysis: A DeepAR and FinBERT Approach With Controlled Seed Sampling

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ABSTRACT

Changes in macroeconomic policies and market news have considerable influence over financial markets and subsequently impact their predictability. This study investigates whether incorporating sentiment analysis can enhance the accuracy of ETF price predictions. Specifically, we aim to predict ETF price movements using sentiment scores derived from news article summaries. Utilizing FinBERT for sentiment analysis, we quantify the sentiment of these summaries and integrate these scores into our predictive models. We employ DeepAR as a probabilistic model and compare its performance with LSTM in predicting ETF prices. The results demonstrate that DeepAR generally outperforms LSTM and that integrating sentiment scores significantly improves prediction accuracy. Given the promising outcomes, we also introduce a fixed “Seed” approach to ensure greater reliability and stability in our probabilistic predictions, addressing the need for robust sampling techniques in practical applications.

1 | Introduction

An exchange traded fund (ETF) is a type of investment fund similar to mutual funds, with the exception that it is traded on stock exchanges. Recent market statistics suggest that ETFs are an important class of investment products. The assets under management (AUM) for open-ended funds worldwide was more than US\$64.63 trillion at the end of March 2021. ETFs accounted for 12.8% of this figure, amounting to an AUM of US\$8.27 trillion. Moreover, about 98% of total ETF assets are index-based instruments. US ETFs had a total AUM of more than US\$6.78 trillion at the end of August 2021 from about US\$2 trillion at the end of 2014 (Saha et al. 2022). Unlike a stock, which represents a stake in a single entity, an ETF contains a diversified portfolio of underlying assets. This composition

makes ETFs a preferred choice for investors seeking diversification (Gastineau 2001).

Macroeconomic policies and public announcements critically influence the daily trading volumes of ETFs (Nadler et al. 2015). Macroeconomic policies in the United States, including alterations in the federal funds rate and shifts in market volatility, not only impact domestic ETF markets but also exert a consequential influence on global financial markets (Yavas and Rezayat 2016).

The significance of machine learning (ML) techniques in forecasting ETF performance is paramount. These methods empower financial analysts and investors to execute data-informed decisions within a dynamic and unpredictable

market context. Utilizing sophisticated algorithms, ML models adeptly analyze extensive datasets, both historical and current, discern complex correlations, and anticipate market directions with notable precision (Fischer and Krauss 2018). ML-driven predictions equip investors with insights to enhance portfolio performance, reduce risks, and seize lucrative trading prospects. Techniques such as deep learning and neural networks facilitate the simultaneous examination of numerous variables, a process essential for comprehending the complex interplay in the ETF market (Ding et al. 2015). Moreover, natural language processing (NLP) enables the sentiment analysis of news articles and social media content, yielding crucial insights into market moods that can influence ETF pricing (Bollen et al. 2011).

The objective of this research is to investigate the influence of integrating news sentiments into the process of ETF prediction, assessing its potential to enhance prediction accuracy using advanced sentiment analysis (FinBERT) and probabilistic prediction techniques (DeepAR).

2 | Related Work

The utility of ML algorithms in ETF and stock prediction has been extensively researched. Specific methodologies such as long short-term memory (LSTM) networks (Chen et al. 2015), random forests (Zhang et al. 2023), and support vector machines (SVMs) (Baek et al. 2020) have been applied to historical market data to project price movements. These techniques have yielded promising results, enriching insights into market dynamics and enhancing the decision-making process for investors.

Advanced deep learning methods, notably convolutional neural networks (CNNs) and recurrent neural networks (RNNs), are increasingly utilized in the realm of stock and ETF prediction. A study compared ARIMA, LSTM, and GRU models for time series forecasting (Yamak et al. 2019) using daily Bitcoin exchange rate data against the US dollar from November 28, 2014, to June 5, 2019. The researchers implemented normalization and log transformation techniques, addressing stationarity and seasonality within the data. Findings indicated that ARIMA outperformed in terms of accuracy and computational time, a result potentially influenced by the selected parameters and the relatively limited data volume. Furthermore, the GRU surpassed LSTM in performance, although RNNs are generally more suitable for larger datasets.

A study developed a big data analytics framework that harnesses hybrid ML algorithms to forecast the direction of daily returns for the SPDR S&P 500 ETF, utilizing 60 financial and economic variables for model training and testing (Zhong and Enke 2019). This research applied deep neural networks (DNNs) and traditional artificial neural networks (ANNs) to both the entire preprocessed dataset and subsets transformed by principal component analysis (PCA). It was found that DNNs trained on PCA-enhanced datasets significantly outstripped the classification accuracy of hybrid algorithms using the raw dataset. Moreover, trading strategies informed by DNN classifications based on PCA data modestly surpassed other strategies, including established benchmarks, in performance. These insights

underscore the importance of feature optimization and data transformation in improving ML model accuracy for predicting stock index returns.

In the study of Sheng and Ma (2022), researchers developed an ML framework to detect spot-futures arbitrage opportunities in the China Security Index (CSI) 300, utilizing historical high-frequency data. Initially, econometric models pinpointed arbitrage chances and constructed ETF portfolios that closely tracked CSI 300's movements. A strategy was then designed, featuring non-arbitrage intervals and unwinding timing indicators, which yielded profitable results in a back-test. For predicting unwinding times, four ML techniques were employed: LASSO, XGBoost, BPNN, and LSTM. Their performances were assessed using root mean squared error (RMSE), mean absolute percentage error (MAPE), and goodness of fit (R^2). The LSTM model was the most effective, with an RMSE of 0.00813, MAPE of 0.70%, and an R^2 of 92.09% across the tested period.

The research detailed in Sun and Zhao (2020) evaluated the efficacy of three ML algorithms—multiple linear regression, random forest, and LSTM networks—in forecasting stock prices. This evaluation was based on the closing price data of a NASDAQ ETF alongside statistical factors. Findings indicated a marginally superior performance of closing price data over statistical factors, albeit not significantly. Multiple linear regression emerged as the most apt for stock price prediction, with random forest as the runner-up, albeit with a tendency towards overfitting. The LSTM network lagged, evidencing the highest RMSE and MAPE values. The study posits that while multiple linear regression holds promise for stock price forecasting, optimal predictive accuracy may require further investigation into the most effective combination of input variables and model configurations.

Kim and Kim (2019) endorse a feature fusion approach for predicting stock prices, integrating different data representations from financial time series. They introduced the LSTM-CNN model, which merges temporal features extracted by LSTM and image features extracted by CNN, using both stock time series and stock chart images. Evaluating this model with SPDR S&P 500 ETF data, they found that the LSTM-CNN model surpassed the performance of single models (CNN and LSTM) in stock price prediction. Moreover, they ascertained that candlestick charts are the most effective type of stock chart image for this forecasting task.

Recently, researchers have utilized NLP techniques to extract sentiment from textual data, gauging market sentiment, and investor emotions. In the study of Li et al. (2014), researchers utilized financial news from Hong Kong encompassing both company-specific and market-related content spanning January 2003 to March 2008. They correlated this data with daily stock selections post sentiment analysis to assess the news impact. Their findings indicated that incorporating sentiment analysis enhances prediction accuracy. Complementary research by Mohan et al. (2019) compiled daily stock prices of S&P500 companies over 5 years and over 265,000 related financial news articles. Employing LSTM for predictions, their results corroborated a robust link between stock prices and financial news content.

An alternative methodology examined the impact of daily aggregated market news sentiment, derived from the Financial News Sentiment scores by TRNA, on stock returns within the DJIA (Allen et al. 2019). By incorporating this sentiment score and its simple moving average (SMA) into the analysis, the study demonstrated that financial news sentiment significantly enhances the traditional asset pricing model and serves as a viable factor in asset valuation.

In related research, sentiment analysis of over 3.6 million Reuters news records was utilized within a vector autoregression model to evaluate its predictive capacity for the Dow Jones Industrial Average stock index returns (Uhl 2014). The findings indicated that Reuters-sourced sentiment offers greater explanatory and predictive value for stock return changes compared to macroeconomic indicators. Notably, negative sentiment from Reuters exhibited a stronger predictive influence than positive sentiment. The study concluded that investment strategies informed by Reuters sentiment data achieve noteworthy outperformance, evidenced by high success rates and Sharpe ratios.

In the research of Sousa et al. (2019), the authors utilized a BERT model fine-tuned on manually labeled financial news to allocate sentiment scores—positive, negative, or neutral. This fine-tuning process involved 582 documents sourced from notable outlets such as CNBC, Forbes, The New York Times, The Washington Post, Business Insider, and others. A comparative analysis of BERT's performance with SVM and TextCNN models revealed that BERT demonstrated superior effectiveness over the CNNs and word embeddings approaches.

Souma et al. (2019) explored the capacity of historical news sentiments to predict stock returns post-news publication. Sentiments were categorized as positive or negative based on the subsequent stock price fluctuations. They used articles from the Wikipedia and Gigaword corpora dated 2014 and applied GloVe (Global Vectors for Word Representation) in a TensorFlow-based deep learning network. The study also analyzed high-frequency data from the Thompson Reuters News Archive and the DJIA 30 Index stocks (2003–2013), using RNNs with LSTM units. A key finding was that predictive accuracy was enhanced when the model was trained on news with the most extreme positive and negative sentiment scores rather than a random selection of news.

The inherent unpredictability of stock price forecasting casts doubts on the dependability of deterministic methods. Consequently, the research community has begun to explore probabilistic approaches that account for uncertainty in future market predictions and portfolio management, acknowledging the limitations of traditional forecasting techniques (Cheng et al. 2023).

In this study (Kim et al. 2023), the authors analyzed summary text data from The New York Times, spanning from January 30, 2018, to December 30, 2022. They applied FinBERT to these concise news article summaries to calculate sentiment scores. These sentiment scores, along with stock price data, were then used to train an LSTM model for stock price prediction. Using error measures such as MSE, RMSE, and MAE, the results

demonstrated that incorporating sentiment scores significantly improved prediction accuracy across all metrics.

3 | Methodology

This research explores the integration of three distinct models: LSTM model, probabilistic deep autoregressive model (DeepAR), and financial BERT (FinBERT), each selected for its unique features and applicability to the study.

3.1 | LSTM

The LSTM network is a specialized type of RNN; in Hochreiter and Schmidhuber (1997), the authors introduced the LSTM cell. They improved the remembering capacity of the standard recurrent cell by introducing a “gate” into the cell (Yu et al. 2019). However, to solve the “output function saturation” issue that original LSTM suffers from, Gers, Schmidhuber, and Cummins (2000) proposed a “forget gate” that enables an LSTM cell to learn to reset itself once their contents are out of date and hence useless, as illustrated in Figure 1.

Therefore, the proposed version of LSTM cell can be expressed mathematically as follows:

$$\begin{aligned} f_t &= \sigma(W_f * h_{t-1} + W_f * x_t + b_f) \\ i_t &= \sigma(W_i * h_{t-1} + W_i * x_t + b_i) \\ \tilde{c}_t &= \tanh(W_{ch} * h_{t-1} + W_{cx} * x_t + b_c) \\ c_t &= f_t * c_{t-1} + i_t * \tilde{c}_t \\ o_t &= \sigma(W_o * h_{t-1} + W_o * x_t + b_o) \\ h_t &= o_t * \tanh(c_t) \end{aligned} \quad (1)$$

where f_t is the forget gate that can have values between 0 and 1; when it equals to 1, it will keep the information; meanwhile, a value of 0 means it gets rid of all the information.

3.2 | DeepAR

The autoregressive RNN model represents a class of probabilistic-based networks, with DeepAR being a specific variant within this category. DeepAR model differs from the above LSTM model, because LSTM is deterministic model, while DeepAR is

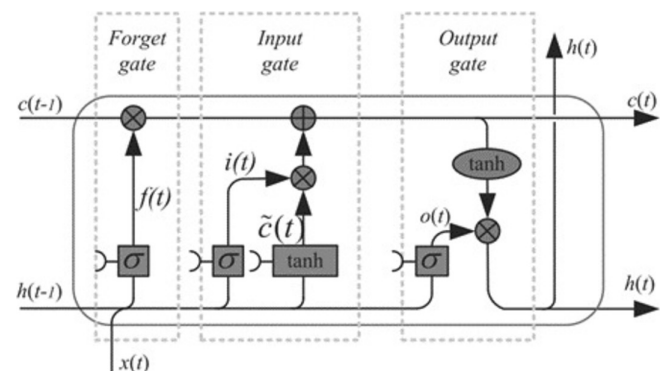


FIGURE 1 | Architecture of LSTM with a forget gate (Yu et al. 2019).

a probabilistic model. The latter may be more appropriate in the context of financial data, given its inherently uncertain nature. DeepAR proposes an RNN architecture for probabilistic forecasting, incorporating Gaussian likelihood for real-valued data and negative-binomial likelihood for positive count data, with special considerations for time series with widely varying magnitudes (Salinas et al. 2017).

During the training process of DeepAR, as depicted in Figure 2, the network inputs at each time step t include the covariates i_t , the target value at the previous time step $x_{i,t-1}$, and the previous network output $h_{i,t-1}$. The network output

$$h_{i,t} = h(h_{i,t-1}, x_{i,t-1}, x_{i,t}, \Theta) \quad (2)$$

is then utilized to calculate the parameters $\theta_{i,t} = \theta(h_{i,t}, \Theta)$ of the likelihood $\ell(x|\theta)$, which is used for training the model parameters. As for prediction, the history of the time series $x_{i,t}$ is fed in for $t < t_0$; then in the prediction range for $t \geq t_0$, a sample $\hat{x}_{i,t} \sim \ell(\cdot|\theta_{i,t})$ is drawn and fed back for the next point until the end of the prediction range $t = t_0 + T$ generating one sample trace. Repeating this prediction process yields many traces representing the joint predicted distribution (Salinas et al. 2017).

3.3 | FinBERT

The bidirectional encoder representations from transformers (BERT) was introduced to improve the fine-tuning process. Unlike previous language models as they predict the next word given previous ones, BERT “masks” a randomly selected 15% of all tokens. With a SoftMax layer over vocabulary on top of the last encoder layer, the masked tokens are predicted. A second task BERT is trained on is “next sentence prediction.” Given two sentences, the model predicts whether these two follow each other (Devlin et al. n.d.; Araci 2019). Yet, to further fine-tune the model for tasks in financial domain, “FinBERT” was introduced by Araci (2019). To achieve the finetuning process, they used two data sources:

- A filtered subset of Reuters’ TRC2 corpus that was refined using financial keywords. The resulting corpus, TRC2-financial, includes 46,143 documents with more than 29M words and nearly 400 K sentences.
- Financial Phrasebank consists of 4845 English sentences selected randomly from financial news found on LexisNexis

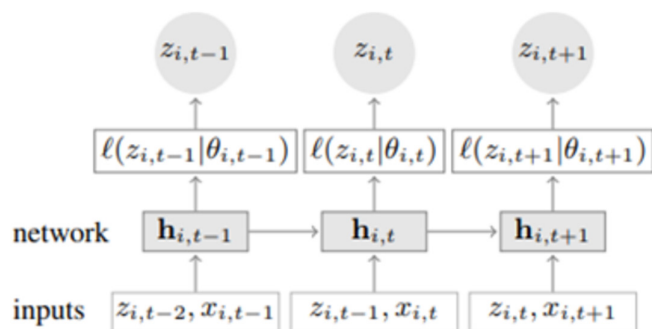


FIGURE 2 | Mathematical operations in DeepAR during training.

database. These sentences then were annotated by 16 people with background in finance and business.

To mitigate catastrophic forgetting during FinBERT fine-tuning, which occurs when the model loses knowledge from the original language modeling task while adapting to the new financial domain task, the authors used the discriminative fine-tuning where they used lower learning rates for lower layers on the network. As lower layers represent the deep-level language information, the upper ones include information for actual classification task. Therefore, they fine-tuned them differently. Finally, authors conduct experiments to benchmark FinBERT against BERT and LSTM, where results show that FinBERT outperforms state-of-the-art ML methods.

In this study, FinBERT is employed for sentiment analysis on relevant ticker news. The outcomes of this analysis are integrated into an “Enriched Dataset,” as illustrated in Figure 3. Further elaboration on this process will be provided in the experiment section for comprehensive understanding.

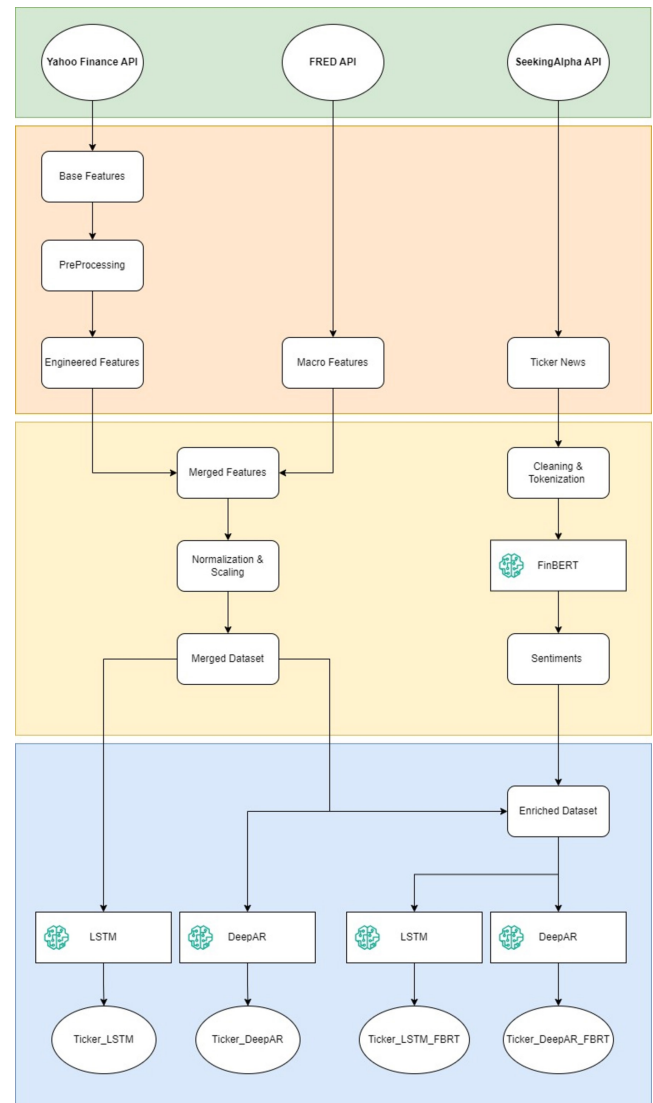


FIGURE 3 | Methodology-driven implementation and data flow analysis with selected models.

4 | Data

The dataset covers eight different US ETFs, each one of them has its own starting date as Table 1 shows. These ETFs have different assets, where BND and GOVT are Bond ETFs offering fixed income securities, GLD is Gold ETF tracking the precious metal's price, IVOO and VIOO are Equity ETFs representing mid-cap and small-cap US companies, respectively, JNK is a high-yield corporate bond ETF, indicating higher risk and return potential, and SPY and VOO: large-cap equity ETFs mirroring the S&P 500 Index, providing broad market exposure.

Moreover, the risk profile is varied from one of these ETFs to another, for example, BND and GOVT are lower risk due to investments in government or high-quality bonds, where JNK represents a higher risk due to investments in lower-rated, high-yield corporate bonds, and finally, IVOO, VIOO, SPY, and VOO are moderate to high risk, reflecting equity investments and market volatility.

The availability of historical data for these ETFs is also different, where SPY, for instance, available since 1994, offering a long-term performance perspective and, on the other hand, BND, GLD, GOVT, IVOO, JNK, VIOO, and VOO, their data starts from 2005 onwards, providing a more recent historical context for performance evaluation as Table 1 shows.

All the eight ETF tickers shared common ticker "base" attributes, such as "Open," "Close," "High," "Low," "Volume," and "Adjusted Price." The data for these features are obtained from the "Yahoo Finance" API, based on ticker own start date.

In order to enhance the model ability to react to the changes in the economic policy decisions, a set of selected US macroeconomic and monetary indicators has been added to the base attributes formulating the "merged dataset," including attributes such as the Federal Reserve Fund Rate, Consumer Price Index, and Federal Reserve Total Assets as Table 2 shows. These indicators have been selected based on correlation test results against the different tickers and based on the recommendations from

TABLE 1 | Selected ETF tickers.

No.	Code	Name	Data start date
1	BND	Vanguard Total Bond Market Index Fund	01/05/2007
2	GLD	SPDR Gold Shares	01/01/2005
3	GOVT	iShares US Treasury Bond	01/03/2012
4	IVOO	Vanguard S&P Mid-Cap 400 Index Fund	01/10/2010
5	JNK	SPDR Bloomberg High Yield Bond	01/01/2008
6	VIOO	Vanguard S&P Small-Cap 600 Index Fund	01/10/2010
7	SPY	SPDR S&P 500 ETF Trust	01/01/1994
8	VOO	S&P 500 ETF	01/10/2010

domain experts as well. The primary source of data for these features is the "Federal Reserve Economic Data" API.

The news data utilized in this study were sourced from the "Seeking Alpha" API. These news articles were pre-tagged based on their corresponding ETF ticker(s). The data collection process involved real-time querying of news articles, with results structured on a daily basis, commencing from the same start day as the respective ETF under consideration.

5 | Experiment

The experimental framework, as illustrated in Figure 3, delineates the distinct stages of data flow and model integration. In this experimentation, three diverse APIs, namely, Yahoo Finance, FRED, and Seeking Alpha, were utilized as primary data sources. The initial phases involved data cleaning and pre-processing. Additionally, calculated features, such as the "3-Day Moving Average," "10-Day Moving Average," "Buy-Sell on

TABLE 2 | Selected macroeconomic indicators.

Code	Name
FEDFUNDS	Federal Reserve Fund Rate
CPALTT01USM661S	Consumer Price Index
DTWEXBGS	Nominal Broad US Dollar Index
WALCL	Federal Reserve Total Assets (QE)
GDPCC1	Real Gross Domestic Product
DGS3MO	Market Yield on US Treasury Securities at 3-Month Constant Maturity
EMVOVERALLEMV	Overall Equity Market Volatility Tracker
RBUSBIS	Real Broad Effective Exchange Rate
PAYEMS	Employment Level
IRLTTLT01USM156N	Long-Term Government Bond Yields: 10-Year
DGS2	Market Yield on US Treasury Securities at 2-Year Constant Maturity
DGS10	Market Yield on Treasury Securities at 10-Year
DFII10	Market Yield on Treasury Securities at 10-Year-Inflation
VIXCLS	CBOE Volatility Index
EMVMACROINTEREST	Interest Rates Equity Market Volatility Tracker
EMVMACROBUS	Investment Sentiment Equity Market Volatility Tracker

Close,” “Buy-Sell on Open,” “Day of the Year,” and “Month of the Year,” were incorporated into the foundational dataset to capture supplementary data trends.

Upon the integration of macroeconomic and base features, data standardization was carried out employing the Standard Scaler and Min-Max Scaler techniques. Subsequently, a correlation matrix analysis was performed to identify and eliminate any correlated features, resulting in the formation of the “Merged Dataset.” This combined dataset was subsequently utilized for evaluating the performance of two distinct models: LSTM, a deterministic model, and DeepAR, a probabilistic model.

To assess the influence of news sentiment integration on the models’ performance, an “Enriched Dataset” was constructed. This involved querying the Seeking Alpha API for related ETF news, followed by data cleaning and tokenization. The news data were processed through “FinBERT” to derive sentiment classes, including “Positive,” “Negative,” and “Neutral”; moreover, as there can be more than record for the same day, a weighted average has been calculated to assign the class. These sentiment features were then appended to the dataset and applied to both the LSTM and DeepAR models.

Subsequently, the target feature, the “close” price, was removed from the dataset. The data were then split into two subsets: 20% was allocated for testing, while the remaining 80% was designated for training. During the training phase, the sliding window technique was employed, with the models utilizing 30 time-steps for training and predicting the subsequent 15 time-steps. The model architecture encompassed a fully connected RNN structure with seven hidden layers and a rectified linear unit (ReLU) activation function. The training phase employed the Adam optimization algorithm with a learning rate of 0.001, spanning a total of 700 epochs. In the case of “DeepAR,” evaluations were conducted for both “50%” and “90%” confidence intervals.

5.1 | Controlled Seed Sampling

“Seed” is a number that is used to initialize the random number generator. The seed determines the sequence of random numbers that will be generated and provides a way to reproduce the same sequence of random numbers across different runs of a program. In DeepAR model, the “Seed” lives in the MXNet C++ API layer.

Giving the business decisions nature that tend to use a fixed values to evaluate risk, and in order to balance the conflicting goals of controlling the randomness of sampling while maintaining a probabilistic approach, we should set a fixed seed. However, to ensure that the seed can still be updated with new input data, it may be appropriate to incorporate both the ticker identifier and the date of the data update into the seed. This can be expressed mathematically as the product of the ticker ID and the sum of the day of the year and the year of the data update:

$$S = T * (doy + y) \quad (3)$$

where S represents the value of the seed, T represents the unique identifier of the ticker, “doy” represents the day of the year, and y represents the year of the data update. By incorporating both

the ticker identifier and the data update date into the seed value, we can strike a balance between controlling the randomness of the sampling process and allowing for changes when new data becomes available.

5.2 | Performance Metrics

In the present study, the MAPE was utilized to assess the accuracy of the time-series models in comparison to actual values.

TABLE 3 | Experimental results performance.

No.	Experiment	MAPE	RMSE
1	VOO_LSTM	0.079	32.84
2	VOO_LSTM_FBRT	0.066	27.69
3	VOO_DeepAR	0.022	12.46
4	VOO_DeepAR_FBRT	0.018	8.06
5	VIOO_LSTM	0.023	2.29
6	VIOO_LSTM_FBRT	0.010	1.21
7	VIOO_DeepAR	0.054	5.03
8	VIOO_DeepAR_FBRT	0.044	4.14
9	SPY_LSTM	0.21	91.80
10	SPY_LSTM_FBRT	0.068	29.49
11	SPY_DeepAR	0.023	10.89
12	SPY_DeepAR_FBRT	0.018	8.31
13	JNK_LSTM	0.0124	1.27
14	JNK_LSTM_FBRT	0.0071	0.73
15	JNK_DeepAR	0.0065	0.69
16	JNK_DeepAR_FBRT	0.0046	0.43
17	IVOO_LSTM	0.043	3.92
18	IVOO_LSTM_FBRT	0.022	1.94
19	IVOO_DeepAR	0.019	2.02
20	IVOO_DeepAR_FBRT	0.011	1.15
21	GOVT_LSTM	0.040	0.88
22	GOVT_LSTM_FBRT	0.034	0.76
23	GOVT_DeepAR	0.022	0.53
24	GOVT_DeepAR_FBRT	0.008	0.15
25	GLD_LSTM	0.069	12.44
26	GLD_LSTM_FBRT	0.044	8.22
27	GLD_DeepAR	0.022	5.80
28	GLD_DeepAR_FBRT	0.011	1.96
29	BND_LSTM	0.0261	1.90
30	BND_LSTM_FBRT	0.0114	0.98
31	BND_DeepAR	0.0075	0.66
32	BND_DeepAR_FBRT	0.0067	0.52

MAPE is defined as the average absolute percent error of each time period, calculated as the ratio of the absolute difference between actual and predicted values and the actual values (scikit-learn.org 2023a). The mathematical expression for MAPE can be represented as follows:

$$\text{MAPE}(y, \hat{y}) = \frac{1}{n_{\text{samples}}} \sum_{i=0}^{n_{\text{samples}}-1} \frac{|y_i - \hat{y}_i|}{\max(\epsilon, |y_i|)} \quad (4)$$

Additionally, the RMSE was employed to evaluate the quality of the predictions. RMSE provides information on the deviation of the predictions from the true values, as determined through the Euclidean distance (C3 AI 2023). The calculation of RMSE involves finding the square root of the mean square error, as described by the following equation (scikit-learn.org 2023b):

$$\text{MSE}(y, \hat{y}) = \frac{1}{n_{\text{samples}}} \sum_{i=0}^{n_{\text{samples}}-1} (y_i - \hat{y}_i)^2 \quad (5)$$

MAPE measures the accuracy of the forecasted values as a percentage of the actual values. RMSE measures the average magnitude of the errors between forecasted and actual values. Lower MAPE and RMSE values indicate better accuracy.

6 | Results and Discussion

The incorporation of sentiment analysis using FinBERT has demonstrated a significant enhancement in the prediction

accuracy of ETF prices. Specifically, models that integrate sentiment analysis (LSTM_FBRT and DeepAR_FBRT) consistently outperform their non-sentiment counterparts (LSTM and DeepAR) across all examined ETFs. This improvement is evident in the reduction of error measures such as MAPE and RMSE. The added context provided by sentiment analysis enables these models to better capture market sentiment, which is a critical factor influencing asset prices. These findings are consistent with previous studies, such as those referenced in Kim et al. (2023).

Moreover, the DeepAR models, both with and without FinBERT (DeepAR and DeepAR_FBRT), generally outperform the LSTM-based models in terms of both MAPE and RMSE. The superior performance of DeepAR-based models can be attributed to their advanced probabilistic forecasting framework, which effectively captures complex patterns in ETF data, as demonstrated in Table 3. Despite the overall trend of improved performance with sentiment analysis, some discrepancies were observed. For instance, the MAPE and RMSE values for VIOO_DeepAR_FBRT were higher compared to VIOO_LSTM. These discrepancies can be attributed to several factors. The VIOO ETF, which tracks the S&P SmallCap 600 Index, may exhibit unique volatility and trading behaviors that sentiment analysis does not fully capture, potentially introducing additional noise rather than clarity. Furthermore, VIOO might be more sensitive to short-term market fluctuations and specific factors that the DeepAR model's probabilistic framework does not address as effectively as the LSTM model.

Overall, the incorporation of FinBERT for sentiment analysis consistently reduces MAPE and RMSE values across all ETFs,



FIGURE 4 | Prediction results: LSTM and LSTM_FBRT models.

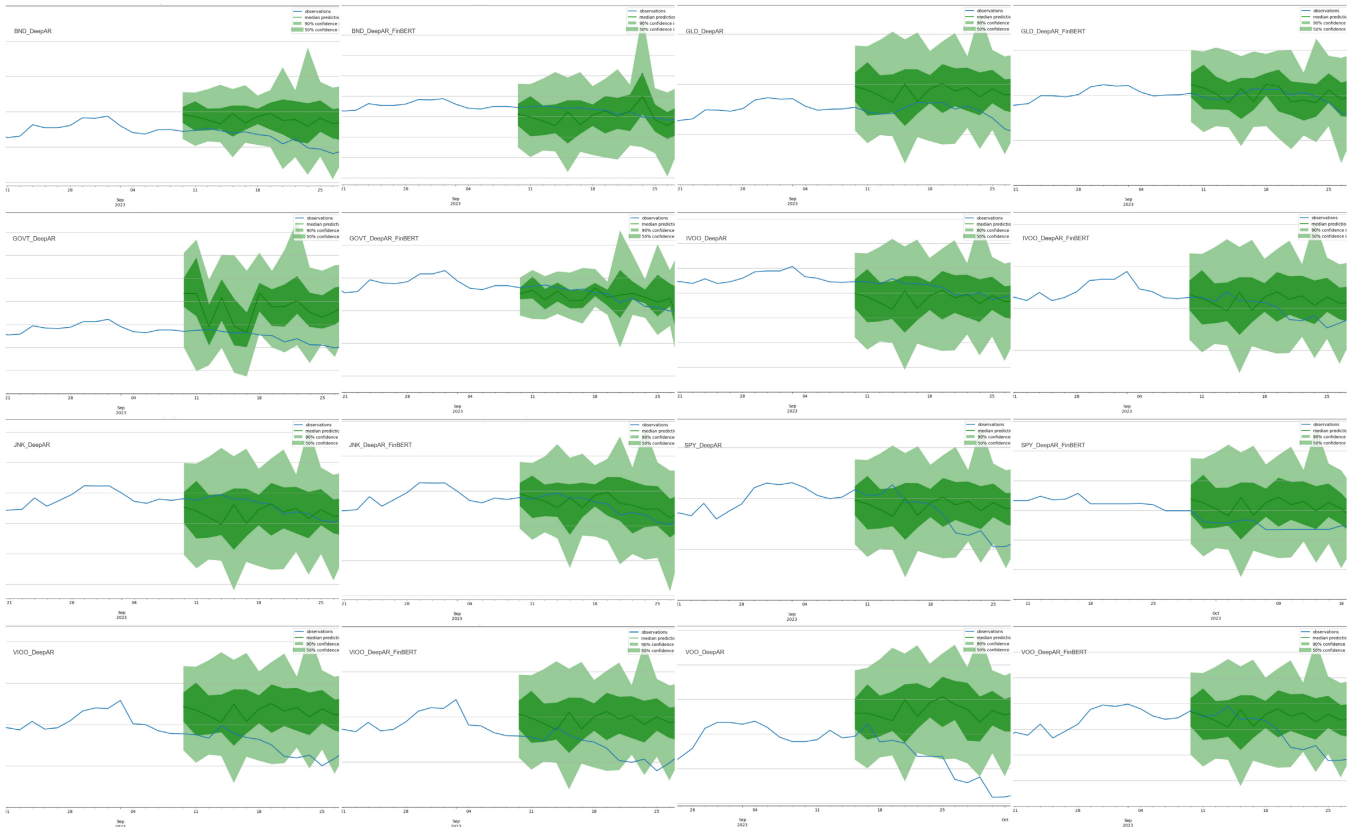


FIGURE 5 | Prediction results: DeepAR and DeepAR_FBRT models.

highlighting the importance of considering market sentiment in ETF price prediction tasks. However, different ETFs present varying levels of prediction difficulty, as indicated by their respective error metrics. For example, BND (a bond market ETF) generally exhibits lower error metrics compared to more volatile ETFs such as SPY (S&P 500) and GLD (gold ETF).

ETFs represent diverse sectors and asset classes, each influenced by different factors. The prices of bonds (BND and GOVT) are affected by different dynamics than those influencing gold (GLD) or S&P 500 companies (VOO and SPY). This diversity necessitates tailored forecasting models for each ETF. ETFs with higher trading volumes and liquidity, such as VOO and SPY, tend to have more stable price movements due to the influence of a larger number of market participants, as illustrated in Figures 4 and 5. Moreover, ETFs in sectors where investor sentiment plays a significant role, such as high-yield bonds (JNK), show notable improvements when sentiment analysis is integrated.

In summary, the superior performance of DeepAR models, particularly when combined with FinBERT for sentiment analysis, underscores the potential of advanced ML techniques in forecasting ETF prices. The variability in results across different ETFs emphasizes the necessity of customizing forecasting models based on the specific characteristics of the assets. Additionally, these findings highlight the critical importance of considering market sentiment, particularly in sectors where investor perceptions and sentiment significantly impact price movements.

7 | Conclusion

In this study, the exploration of integrating news sentiments to enhance ETF price prediction accuracy was undertaken. Three distinct models—LSTM, DeepAR, and FinBERT—were carefully incorporated, each selected for its unique strengths, illuminating the nuanced influence of market sentiment on asset price forecasting.

To conduct the research, data were sourced from diverse channels and processed with specific features using APIs. The models' performance was evaluated using metrics like MAPE and RMSE. The analysis consistently demonstrated the superior performance of LSTM_FBRT and DeepAR_FBRT compared to their counterparts. This highlighted the significant impact of FinBERT's sentiment analysis in understanding market sentiment's crucial role in shaping asset prices.

Additionally, a fundamental enhancement was introduced—implementing a fixed “seed” approach in DeepAR. This innovation ensured stability and reproducibility of results across experiments, maintaining a consistent sequence of random numbers for precise control over sampling processes. This precision in controlling sampling processes is vital for seamless business implementation, ensuring consistent and reliable results. Furthermore, our investigation yielded valuable insights into the diverse behaviors of different ETFs. Some assets were inherently more predictable due to lower volatility and stable trading, while others presented challenges due to their market

volatility. These varying complexities emphasized the necessity of custom-tailored predictive models, designed to accommodate the unique traits of individual assets.

In summary, this research underscored the transformative potential of advanced probabilistic models such as DeepAR, especially when enhanced with sentiment analysis tools like FinBERT. It emphasized the importance of tailoring prediction models to the unique characteristics of each asset and highlighted the critical role of market sentiment in accurate price forecasting. For future work, conducting stress tests and scenario analyses is essential to evaluate the resilience of predictive models under different market conditions and identify potential vulnerabilities.

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Data Availability Statement

Research data are not shared.

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